#systemmapping

Guided by an explanatory challenge, develop simplified representations of complex systems by conceptualizing their components and relating them to one another.

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Defining the HC

Pitch

A system might seem vast, too complex, and too dynamic to analyze and fully understand. But as with any unexplored territory, the right map can show us the way to our final destination.

Paragraph Description

According to Plato, there is no obvious way to "carve nature at its joints." When mapping complex systems, we make choices about how to represent reality. Just like a mapmaker chooses to represent geographical reality by highlighting rivers, valleys, or national borders rather than religious borders, the way we dissect systems depends on, and should be guided by, our purpose. In empirical research, our purpose is addressing the "explanatory challenge." Complex systems can be deconstructed in a variety of ways that define both the scope of the system and the relevant attributes of its components (in social systems, most commonly, people). Thus, the key part of deconstructing a system is conceptualizing its constituent parts in several ways in order to choose or synthesize the “mapping” of the system that is most appropriate for addressing the explanatory challenge. For example, in analyzing a football team one might think of players in terms of their position on the field, their running speed, or their personality types. Depending on the question at hand, one may or may not include the players’ personal trainers, the team’s management, or even the fans. One specific application of this concept is considering organizational structure: there are various ways in which the same group of people and functions may be related to one another, such as a strict hierarchy, a matrix, or a flat structure.

Categorization

Habit of Mind | Complex Systems | Interacting Effectively | Interacting with complex systems

Cornerstone Introduction

Class: Complex Systems | Semester One

Unit: Characteristics of Complex Systems

Big Question: How should the world be governed?

General Example

You work at a hospital that’s trying to reduce spending. Your manager asks you to make a diagram showing the structure of the organization to identify inefficiencies. You are contemplating how to create this diagram. You first choose to represent the organization by separating it into different departments because the employees within each work together closely, and therefore are expected to share some norms and views about the organizations. While this decomposition reveals the dynamics of interactions between large groups such as the lack of communication between the medical staff in different departments or between the medical and cleaning staff, there are no obvious inefficiencies. You then decide to deconstruct the system by focusing on the roles of individuals and find that each department has its own educational campaigning function and that there is little communication between those working on educational campaigns in different departments. It turns out that there is a large amount of money being spent on educational campaigns overall. This was not obvious when the system was deconstructed by department because education campaigns are funded by each department separately, at relatively low cost. You advise making educational campaigns a central process, consolidating common costs and being able to prioritize what should be budgeted.

Footnote: *The explanatory challenge was identified: reduce spending. To understand the problem and find possible solutions, the system was deconstructed in several ways and the chosen deconstruction is the one that addresses the explanatory challenge. In this case, deconstructing by function allowed us to identify something that was replicated in each department, and thus driving an overall inefficiency in the system.*

Is it this HC?

Is #systemmapping the right HC? If the answer to the following questions is yes, #systemmapping could be useful:

1. Have you considered multiple grouping for the agents or components and explained which decomposition is best suited to answer your problem?
2. Do the distinct properties of the agents/components matter for your explanatory challenge?
3. Are you comparing different ways to decompose agents/components that operate on the same scale?
4. Are you deconstructing the system into functional units and exploring the idea of interactions among the different components to explain the overall behavior or outcomes?

Depending on the work product, there might be other HCs to consider. Not all deconstructions are system mapping.

	The following HC might apply if the work:
#networks	• Focuses on the connections between agents, rather than the differences among agents.

	• Identifies, describes, or evaluates the connective structure of nodes in a system and uses this to address the explanatory challenge.
#emergentproperties	• Describes the collective influence of agents/components, for which a system mapping might be useful.
#levelsofanalysis	• Focuses on the unique benefits of shifting perspectives between levels of analysis, zooming in or out to provide new or unique insights whereas #systemmapping is more general about conceptualizing and deconstructing a system.
#breakitdown	• Deconstructs a problem into subproblems. • Identifies tractable sub-problems that can be explored in isolation, then reassembled into a solution.

Self Study

Try answering the questions on your own before checking the example answers.

👆 What questions can you ask yourself when considering multiple system deconstructions?

What different types of deconstructions are common for the system I am analyzing? Could multiple different deconstructions be valuable for the explanatory challenge? What are the tradeoffs of the different deconstructions? What new insights do each provide? 2

👆 **Familiarizing yourself with organizational design may inform you on the interactions and impact of individual agents. What are some of the common approaches of organizational design? List at least 3. Refer to the "Organizational Structure in Complex Systems" (see Key Resources section) guide for relevant information.**

Hierarchical Organizations, Divisional/Business Unit Hierarchy, Geographic Hierarchy, Matrix Organizations, Flat/Organic Organizations, Circular Organizations

👆 **What are the main requirements for a strong application of #systemmapping?**

A strong application of this HC requires (1) a crisp definition of the system to be mapped; (2) a clear statement of the explanatory challenge; (3) an analysis of the various possible decompositions; and (4) a concise explanation of the deconstruction chosen.

👆 **Imagine that you want to create a system map of a school to analyze why students in their sophomore year put more hours on average into their daily studies than do students in their senior year. Would the principal of the school be a relevant component to your system? Why or why not? List a few arguments and compare them to the answer key.**

Yes, the principal is relevant.

- The principal has more of authority in the eyes of sophomores than in the eyes of seniors (you would need to explain a causal mechanism in an actual application) 2

- The principal's program tends to put more pressure on younger students because it assumes that they would maintain the same level as they move up grades without the required effort of teachers.

No, the principal is not relevant.

- The principal might not be relevant because they have been known to have a more laid-back approach and not put too much pressure on students.
- The principal might not be relevant because they have very little involvement with the direct experience of students, but deal mainly with administrative work.

👆 **Identify what the student did incorrectly in his analysis of the following case-study.**
Case study: **A business student wants to analyze an IT company's marketing division to understand why a significant percentage of employees quit between their 12-th and 18-th month. They include the following agents in their map: the team manager, the marketing branch manager, the CEO, major clients of the company, and the law interns. The student draws the map with a one word explanation above each connection between the agents and writes the following sentence: "Based on the increased working pressure, which is not justified by a reasonable increase in pay, and the worsened work conditions marketing employees between their 12th and 18th month on the job are most susceptible to quitting."**

Based on the description above, the student did not do two main things: did not provide a significant justification for every component's role in the system and did not consider alternative deconstructions. A proper system map requires an analysis of the pros and cons of the deconstruction method and a clear explanation of every agent's relevance which contributes to the understanding of the explanatory challenge.

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Applying the HC

Guided Reflection

- Have you clearly articulated an explanatory challenge?
- Have you deconstructed the system into plausible components? Did you include all agents or components that might be relevant to your explanatory challenge?
- Have you made a visual map of the system, does it help you understand it holistically?
- Have you ensured and explained how the deconstruction is useful to address the explanatory challenge?
- Have you explained how the deconstruction offers insights about the system's behavior?
- Have you provided a clear and reasonable justification for all the agents you chose and the way you grouped them?
- Have you outlined the role and impact that each agent plays in the system?
- Have you considered alternative deconstructions, justified why your deconstruction is better than alternative approaches, and stated why other approaches are inappropriate?

Common Pitfalls

- The application does not include a crisp definition of the system to be mapped.
- The application's explanatory challenge is unclear and/or unstated.
- The application does not consider all relevant agents.
- The application does not explain the relevance of the system deconstruction.
- The application does not provide sufficient detail for each agent/component or omits a part relevant to the subsequent analysis that is not intuitive.
- The application does not justify the effectiveness of the decomposition by comparing it to alternative approaches.
- The application uses system mapping for purposes more relevant to other HCs, like #breakitdown, #networks, and #emergentproperties (refer to section 'Is it this HC?'). 2

Applying the HC at Minerva

This HC is ubiquitously applicable across many fields. Consider these general use-cases:

- You can compare different types of system deconstructions that are common in the relevant discipline, or the different types of organizational structures that are applied to a system.
- You can suggest a new type of deconstruction and compare it with existing ones. You can also combine from common deconstructions in an original way to map the system based on evaluated strengths and weaknesses.
- It works especially well if the different components of the system are complex themselves, or have many possible characteristics to highlight (humans, or departments are a great example), but it can also work in natural sciences, for example grouping species in an ecosystem by their niche or ecosystem service.

Here are some brief examples:

- This HC is relevant in Business classes when analyzing the organization of a company guided by a specific question (e.g. a case study). You should definitely take a look at the different organizational structures discussed in the Minerva guide, "Organizational Structure in Complex Systems" (listed in the key resources below).
- This HC can be particularly useful in the Political Science track when tackling case studies. It can be helpful in deconstructing the structure of a particular sector of the economy (e.g., the civil sector) or of a bureaucracy, where the agent of interest operates.
- There are several use cases throughout the natural sciences.
    - In the physical sciences, when analyzing the motion of multi-component systems, we need to identify the system of interest and the external forces acting on it. The choice of decomposition and systems of interest will depend on which forces or interactions are being examined. Identifying all forces on a particular system of interest is crucial for an accurate model.
    - In natural sciences, when analyzing biological or ecological systems. For example, we could map the species of an ecosystem in various ways, grouping them by trophic level, size, or taxonomic group, depending on the explanatory challenge. The same goes for the internal organs of an

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animal, or parts of a cell or biochemical mechanism. In these fields, try to think of the different existing ways that scientists break down these systems and think whether they suit your problem, or if you need to look beyond and become creative!

Here are some more elaborate examples:

👆 SS / Research Papers for Cognitive Sciences

Example: You want to gain a physiological understanding of aggressive behavior, and you decide to look into the human brain and the interactions in it, focusing on hormones that send messages. You dive deeply into the literature and discover that there are five main areas of the brain that have consistently been correlated with the display of aggressive behavior, so you decide to include them in your system map. However, they do not act by themselves. They interact with other parts of the body, so you begin expanding your map. First you identify that testosterone is considered the responsible hormone for aggression, and so you justify the inclusion of some brain areas through their direct interaction with testosterone. This hormone is not produced in the brain so you find it necessary to also include the body parts that do produce it and consider how factors, such as specific genes, could affect how much testosterone those organs produce. Additionally, you identify neurotransmitters that are related to aggressive behavior in some ways (e.g., dopamine, adrenaline, etc.), and you include their originating centers, as well making sure to explain how the chemicals affect the specific brain parts (as appropriate for your audience). In Cog-sci courses you tend to stay on the same level when mapping a system, in this case the physiological. However, if there is a relevant part that only gets activated through perception of external stimuli then you should include it and explain its connecting mechanisms as well. With such a complicated topic, you should always refer back to your explanatory challenge and, at the end when your map is completed, make sure to go over it again and verify that you have only added components of the required level of specificity and detail - while neurons as the cell base of the brain are definitely related, they may

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be a redundant component, because they are on a different level and you are not interested in individual neurons anyway.

Good when: #systemmapping can help us understand anger from different explanatory lenses. It is important that we compare the way in which we break this system down, for example by giving clear arguments for including or excluding parts of the system. Another thing we could easily do is give an overview of the existing paradigms around anger in scientific literature, and evaluate the strengths and weaknesses of different breakdowns: which parts of anger do they explain effectively; what do they overlook?

👆 AH / Research Papers for Literature

Example: To understand the *Lord of the Rings,* we created a diagram of the characters in the story. We considered different groupings depending on what we wanted to understand from *The Lord of the Rings*. To answer why the story became popular in Japan and not in Korea, we grouped characters by Hofstede's cultural dimensions. The grouping allowed us to entertain more questions. If villains behaved more like Koreans, would the story confirm Japanese bias against Korean values? If most characters behaved like Japanese people, would Japanese readers like the characters more? Is this supported by Cialdini's research that whether you like a person depends on whether you relate to the person? Would this imply that Cialdini's liking principle also applies to fictional characters?

Good when: We stay close to our explanatory challenge, which is about *Lord of the Rings* in the light of the colonial past of Japan and Korea. We should explain why the Hofstede cultural dimension mapping is a relevant tool to divide characters of the cast as opposed to an alternative, like defining them by the MBTI (16 personalities). We should also be clear about what exactly our system is: is it the viewers of LOTR responding to the characters (in which they are also agents of the system), or are we analyzing the behavior of the characters themselves, contextualized in the ideas around good and bad for Koreans and Japanese audiences? 2

👆 IL / Preparing for Civitas

Example: Three students arrive at Civitas, but each student has different intentions:

1. wants to land a job in her dream industry: material sciences.
2. wants to befriend interesting people.
3. wants to understand the challenges in her current rotation city.

The day before the event, the three students get a slide deck showing the people who will be attending Civitas.

The first student goes on LinkedIn and groups people by their connections to material sciences. She defines what kind of job they have (how similar is it to her dream job) and the way in which they relate to other parts of the community that is attending. This way, she understands how to optimize for talking to those most connected. She could also consider the size of the company they work at. What she does well is tailoring her breakdown to her goal, however, it is very time-consuming, is it the most effective way? Or would there be a simpler alternative?

The second student considers two alternatives: looking at the "Fun Facts" section of each slide, grouping people into categories like "weird," "cool," "confusing," and "friendly," and she decides to focus on the people she finds friendly. Or she will pick people at random. One thing she might consider is how the first breakdown simplifies people and could actually undermine her goals. For example, some of our best friends tend to be the people we find most different from ourselves at first, which means we can learn a lot from them and stay excited to discover more about them.

The third student groups people by how involved they are in the city. As a proxy, she considers how many years the person has spent in the city. What this student does really well is optimizing for the effort that goes into the breakdown, if this information is easily accessible. A disadvantage might be that it is not as accurate, since an upper-class person in San Francisco might know very little about the problems in the Tenderloin, unless they are for example 2 a community worker, or civil servant. She could consider breakdowns by function they have in the city, and whether they work with communities.

👆 Business

Example: Sometimes, you could find already synthesized information that you consider a great start for creating a system map for an explanatory challenge you want to address. Let's assume that you want to invest money into stocks of a company that produces **electric vehicles.** You want to know which company currently dominates the market and which one/s could potentially stay on the top or take over in the following years. You find this graph online:

Strong when: consider including other agents that might provide clarity and insights beyond market share: main customers based on the country where the company is registered, as the local legislation might have an impact on corporate policy, consumer groups from different locations, their opinions, etc. Once you have defined the agents based on research and analysis, consider how you could represent them visually on your map, as that may expand your perspectives. One potential way to do it is by placing

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the companies over their respective countries and scaling the size of their symbol to represent their market size. Alternatively, the main priority might be their biggest consumer markets or investors. By drafting and comparing multiple iterations, you will eventually land on the most helpful system map to analyze which stocks will be the most profitable.

Applying the HC Outside of Minerva

#systemmapping is a useful to to break down large systems with interactions that you will come across in daily and professional life, in hobbies and in society. Some examples are:

- **Policy:**How does mapping the economic structure of a city help us track the flow of money?
- **Biology:** How do biotic and abiotic factors affect population growth within a community of animal and plant species?
- **Reading:** How does mapping the vibrant world created in a book or movie help us understand the motives of its characters and analyze the hidden ideas in the artwork?

Whether it is the economic structure of a city and we are trying to track the flow of money; the biotic and abiotic factors that affect population growth with a community of animal and plant species; or the vibrant world created in a book or movie that we are trying to untangle to understand the motives of its characters and analyze the hidden ideas in the artwork.

Here's one specific example:

- Let's assume you aim to reduce your spending. You've realized that people you follow on Instagram have an effect on your spending habits. You ask yourself, "Which people can I mute to reduce how much I spend eating out?" You consider two alternatives:

  A: Grouping users by how frequently they post about eating out.
  B: Grouping users by how often you see their posts.

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  - Choice A helps you identify the source of the eating out craze. Even though you might not look at some user's

posts, people whom you follow might look at their posts. This decomposition helps you understand why posting about eating out has evolved the way it has.
  - Choice B might give you less perspective about eating out posts and more perspective on how different users affect your behavior. Even though a user never posts about eating out, how is their content affecting your mood, and ultimately, causing you to spend money on eating out?
  - Choice A is interesting: you can analyze the factors shaping how eating out is portrayed on social media. But you select Choice B because it is more conducive to reducing your spending habit by paying attention to whom you follow on social media.

Here is another example:

- This example stems from an M25 summer science research project that studied a new porous fence model being set up in a wildlife reserve in Kenya. The fence was intended to allow all animals to cross except rhinos, which were supposed to be enclosed by it. It was important to understand the animals' reactions to the fence, for which they were scored through camera trap footage. With this information, a framework was created to understand how the fence would affect different animals, based on specific intrinsic and extrinsic factors.
  - Extrinsic factors (e.g. fence permeability, height, visibility) include the fence's properties and design.
  - Intrinsic factors (e.g. habitat preferences, species-specific behavior) incorporate the species' characteristics.

  Intrinsic factors influence the way a species reacts to its environment, and in turn, how it reacts to extrinsic fence factors.

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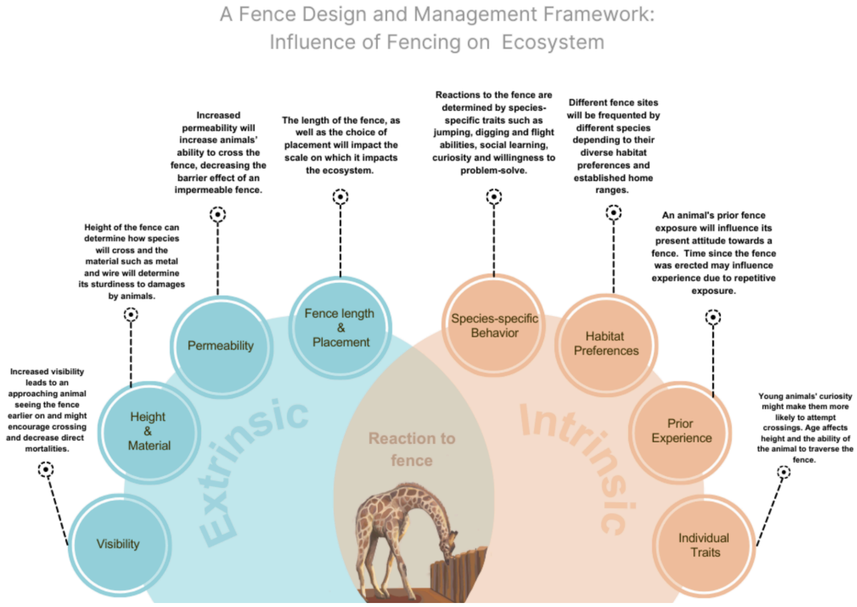

- This framework is of particular interest to reserve managers evaluating the erection of a fence based on factors such as the reserve's habitat, species distribution, and main species of concern. For example, extrinsic factors like fence length and placement can influence habitat that is accessible to animals. A fence that blocks a common migration route or is a hotspot for certain species will have more impact on the animals and ecosystem than one that doesn't. The choice of placement might be a crossing often frequented by elephants. If the fence disables them from crossing, refraining these megaherbivores for grazing in diverse areas will have larger impacts on the ecosystem.

Key Resources

- Sheskin, M. (2021). Characteristics of Complex Systems. https://my.minerva.edu/academics/learning-resources/hc-resources/cornerstone-custom-hc-guides/
  - Brief Introduction: this reading is the overview of how six characteristics of complex systems interrelate. It goes more into depth in the different HCs and how they are different from each other. Find it with all the custom HC whitepapers on this myMinerva page.

- McAllister (2019). Organizational Structure in Complex Systems. https://my.minerva.edu/academics/learning-resources/hc-resources/cornerstone-custom-hc-guides/ 2

  - This reading provides a brief introduction to the different aspects of complex systems. Find it with all the custom HC whitepapers on this myMinerva page.
- Mapping society for a more meaningful world. https://www.youtube.com/watch?v=hAFclNJ--Gs
  - This TED talk provides a great example of the usefulness of system mapping and some questions to ponder on.
- Living Guide to Social Innovation Maps. System Mapping. https://mars-solutions-lab.gitbook.io/living-guide-to-social-innovation-labs/seeing/understanding-the-problem-systems-and-complexity/systems-mapping
  - A webpage with a brief description of the concept and some good visual examples.

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